

Professional Experience

- Feb 2026 **Postdoctoral Fellow**, *Mathematics and Statistics Department*, Indian Institute of Technology Kanpur, mentor: Dr. Bidyut Sanki.
to present
- Sept 2025 **Visiting Scientist**, *Stat-Math Unit*, Indian Statistical Institute Kolkata.
to Feb 2026

Education

- Aug 2020 **Ph.D. in Mathematics**, *Indian Institute of Science Education and Research*, Mohali, Thesis title: ‘Reciprocity of Discrete Subgroups of $PSL(2, \mathbb{C})$ ’.
to Jul 2025 Advisor: Prof. Krishnendu Gongopadhyay
- Jul 2018 **M.Sc. in Mathematics**, *Institute of Science*, Banaras Hindu University, Varanasi.
to May 2020
- Jul 2015 **B.Sc. (Hons.) in Mathematics**, *Ananda Chandra College*, University of North Bengal, Jalpaiguri.
to May 2018

Research Interests

My research interest lies in the broad area of Geometry and Topology, particularly in Fuchsian Groups, Mapping Class Groups, Teichmüller Space, and Hyperbolic 3-Space.

Workshops & Conferences

- Riemann surface and Discontinuous groups 2025 (University of Osaka, Japan).
- Geometry and Analysis of Minimal Surfaces (ICTS, Bangalore, India).
- New Trends in Teichmüller Theory (ICTS, Bangalore, India).
- 20 years of Anosov representations (MPI Leipzig, Germany).
- Young Topologists Meeting, 2024 (University of Münster, Germany).
- Computational Aspects of Thin Groups (NUS, Singapore).
- St. Louis Topology Conference, 2024 (online).
- Int. Colloquium on randomness, geometry, and dynamics, 2024 (IISER Pune).
- Geometry, Analysis and Mathematical Physics Workshop, 2023 (NISER Bhubaneswar).
- NCMW Groups and Computation, 2023 (Krea University).
- Triangle Group Programme, 2021–22 (Bhaskaracharya Pratishthana).
- Inter IISER-NISER Math Meet, 2022 (IISER Kolkata).

- CIMPA School on Geometric Structures, 2022 (BHU).
- Beyond Hyperbolicity and Artin Groups, 2021 (Ohio State University, online).

Talks

- “Torsion Free Subgroup of Triangle Groups”, Bhaskaracharya Pratishthana (online).
- “Reciprocity in the Hecke group”, YoMathS-2023, IISER Mohali, May 2023.
- “Reciprocity in the Hecke Groups”, Inter IISER-NISER Math Meet, Jul 2023 (NISER Bhubaneswar).
- “Reciprocal Geodesics of Hecke groups”, Presidency University, Oct 2023.
- “Combinatorial Growth of Geodesics in Hecke Groups”, YoMathS-2024, IISER Mohali, May 2024.
- “Combinatorial Growth of Reciprocal Geodesics in Hecke Groups”, Krea University, Jul 2024.
- “Combinatorial Growth of Reciprocal Geodesics in Hecke Groups”, NUS Singapore, Aug 2024.
- “Introduction to Anosov representations in rank one cases”, MPI Leipzig, Oct 2024.
- “Reciprocal elements in Picard group”, ICTS Bangalore, Mar 2025(short talk).
- “Reversibility in the fundamental group of Seifert-fibered spaces”, The 6th International Conference *Groups and quandles in low-dimensional topology*, Jun 2025(online).
- “Combinatorial Growth of the reciprocal classes in the Hecke groups”, *Buildings 2025: A celebration of Bernhard Mühlherr’s 60th birthday, UCLouvain*, Sept 2025(online).
- “Reversibility in the groups of three-dimensional spaces”, *Kanazawa Topology Seminar*, Kanazawa University, Japan, Jan 2026.
- “Reciprocal Geodesics and its Recent Developments”, *Riemann surface and Discontinuous groups 2025*, University of Osaka, Jan 2026.

Publications & Pre-prints

1. D. Das and K. Gongopadhyay, “Reciprocity in the Hecke Groups”, *New York J. Math.*, 30 (2024), 722–744.
2. D. Das and K. Gongopadhyay, “On the z-classes in Fuchsian groups”, *Indian J. Pure Appl. Math.*, 2024. DOI:10.1007/s13226-024-00729-7.
3. A. Das and D. Das, “Reversible and other generalised torsion elements in Seifert fibered groups”, arXiv:2403.01541 (2024). (accepted for publication in *Osaka J. Math.*)
4. D. Das and K. Gongopadhyay, “Spherical Growth of Reciprocal Classes in the Hecke Groups”, arXiv:2411.00739 (2024).
5. D. Das and K. Gongopadhyay, “Asymptotic growth of the number of Reciprocal Classes in the Hecke Groups”, arXiv:2506.08449 (2025).
6. D. Das, K. Gongopadhyay and K. Mishra, “Combinatorial Structure of Inert and Ambiguous Classes in Modular Group”, arXiv:2602.19176 (2026)

Teaching Assistant

- MTH201 Curves and Surfaces (2 times), IISER Mohali.
- MTH202 Probability Theory(3 times), IISER Mohali.
- IDC207 Number Theory and Cryptography, IISER Mohali.
- MTH202 Elementary number theory, IISER Mohali .
- AFS-2 Topology-2, Central University of Punjab, Bhatinda.

Invitation/Academic Visits

- April 7-28, 2025 **Academic Visit to Prof. Pralay Chatterjee, IMSc Chennai, India.**
- May 30- Jun 01, 2024 **Academic Visit to Dr. Soumya Dey, Krea University, India.**
- Jan 25- Feb 2, 2026 **Academic Visit to Prof. Hideki Miyachi, Kanazawa University, Japan.**
- May 19-26, 2026 **Academic Visit to Dr. Debanjan Nandi, ISI Kolkata, India.**

Activities

- Aug 2024 to May 2025 Graduate Student Group member, IISER Mohali.
- May 2024 Part of ISI Entrance Exam invigilation, Mohali.

Awards & Grants

- 2018-2020 CSIR-UGC NET qualified (Dec 2018, Jun 2019, Jun 2020).
- Feb 2021 GATE qualified.
- May 17-19, 2024 Support(not availed) from St. Louis University to attend ‘St. Louis Topology Conference’, St. Louis, USA.
- Jun 03–14, 2024 Travel-Grant from IISER Mohali to attend ‘Computational Aspects of Thin Groups’, NUS, Singapore.
- August 5-9, 2024 Travel-Grant from IISER Mohali and University of Münster to attend ‘Young Topologists Meeting 2024’, Münster, Germany.
- Oct 07-11, 2024 Travel-Grant from Clay Mathematics Institute and MPI-MIS to attend ‘20 years of Anosov representations’, MPI-MIS, Leipzig, Germany.